

Evaluating Reinforcement Learning Algorithms for Portfolio Trading

A Reinforcement Learning Study on the S&P 500 ETF (SPY)

Fiyinfoluwa Akano
URN 6962514 · MSc Artificial Intelligence

Supervisor: Dr Cuong Nguyen
University of Surrey

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The problem

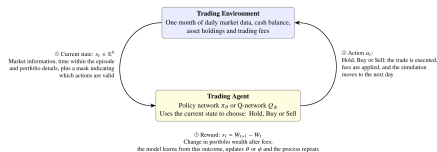
An investor holding a stock-index fund faces the same decision each day: buy more, sell part of the position, or do nothing.

- **Question 1 — performance.** Can reinforcement learning make these decisions better than buying and holding, after transaction costs?
- **Question 2 — behaviour.** Does the learned agent actually use its market information? A market that mostly rises rewards constant buying, so a profitable agent has not necessarily learned anything.
- **Question 3 — data.** Is the historical record of one asset large enough to train these networks at all?

Three algorithms compared: REINFORCE and Deep Q-learning (DQN), implemented from scratch, and PPO from a library. The dissertation answers with **two studies**: real data first, then corrected data.

The trading environment

- Monthly episodes of daily SPY bars (18–23 trading days), \$10,000 starting capital, any open position sold on the final day.
- **State:** nine features — five market (return, momentum, volatility, trend gap, range) and four portfolio (cash, exposure, unrealised PnL, episode clock).
- **Actions:** hold, buy, or sell a fixed $0.25 C_0$; invalid actions masked.
- **Reward:** change in wealth after a 0.05% fee. Risk-aware variant subtracts $\lambda C_0 \Delta dd_t$.



How the evaluation earns trust

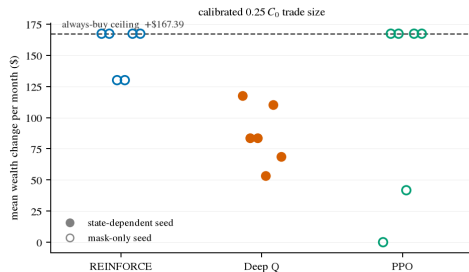
- **Chronological splits.** Train 2018–2022, validate 2023, test 2024–2025. No shuffling, so no look-ahead leakage.
- **Validation-only decisions.** Trade size, risk-penalty weight, and fee checks were all selected on validation data. Every test set was touched once.
- **Six seeds per configuration**, spread reported next to every mean. One good run proves nothing.
- **21 verification tests.** They caught a benchmark bug worth \$38.95 per month — enough to flip the headline conclusion.
- **State-dependence test.** A policy that picks different actions at steps with the *same action mask* must be reading its state. Reported per seed, next to the returns.

The last test separates a policy that responds to the market from one that follows a fixed pattern — the returns alone cannot.

First study: real data alone — a negative result

Trained on the 60 real months of 2018–2022, tested on 2024–2025.

Policy	ΔW (\$/mo)	State-dep.
Buy-and-hold	+180.25	—
Always-buy	+179.75	—
REINFORCE	+171.51	0/6
PPO	+127.35	0/6
Deep Q-learning	+74.68	6/6



No agent beat buy-and-hold. The methods that earned the most ignored their state on every seed; the only method that used its state earned the least. The contradiction needed an explanation.

The diagnosis: the data, not the algorithms

- **60 episodes against 18,000 parameters.** A network that size can memorise every training month; nothing forces it to learn a rule that transfers.
- **The months lean upward.** 29 of 60 training months rose more than 2%; only 16 fell. Under that distribution, always-buy is a nearly optimal answer that needs no state information — the same failure as a classifier trained on a 99%-one-class dataset.
- **Why DQN was worst.** Value estimation needs many visits to similar states under a stable distribution. Sixty shifting months provide neither, so DQN conditioned on noisy estimates.

None of this can be tested on the real data, because the real data is the problem. Testing it needs control over volume, balance, and conditions — which is what the main study builds.

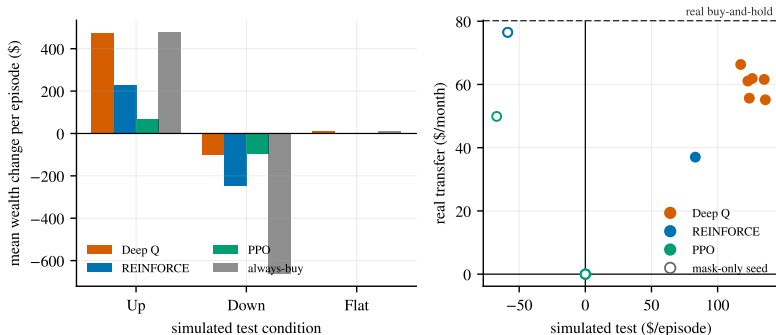
The main study: controlled data, right-sized model

- Market generator with three conditions — up, down, flat — calibrated on the 60 real training months *only*.
- 3,000 balanced training episodes, 300 validation, 600 test. About 60,000 transitions per pass.
- Network cut to 64 and 32 units: $\approx 2,800$ parameters. Data now outnumbers parameters $\sim 20:1$.
- On the balanced test set, **no fixed pattern profits**: buy-and-hold loses \$60.29 per episode; never trading earns \$0. Any profit requires reading the state.

	Months	Drift	Vol.
Up	29	+0.244%	0.97%
Down	16	-0.358%	2.00%
Flat	15	+0.012%	1.10%

Fitted daily parameters per condition; generated ATR matches the real mean within 4%.

Main result: the ranking inverts



- **DQN: +\$126.94 per episode** (seed std \$7.30), all six seeds profitable and state-dependent. It keeps 99% of the up-market profit and cuts the down-market loss by 84%.
- REINFORCE and PPO collapse to corner policies — never trade or always buy. One REINFORCE seed in six found the conditioned policy (+\$83.13), so it is reachable but not reliably found.

The result survives its assumptions

Risk-aware reward ($\lambda = 0.25$, selected on validation):

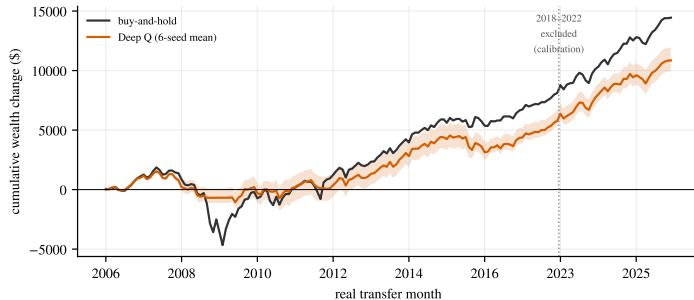
- Down-market loss falls 73% ($-\$87.28$ to $-\$23.26$) for a 1.5% cost in mean return; Sharpe rises 0.285 to 0.332.
- The agent holds less *when the market falls*, not less overall — the selective behaviour the first study could not produce.

Transaction fees swept 0 to 50 basis points:

- DQN stays profitable at every level: $+\$95.27$ per episode at 50 bps, ten times the study's assumption. Buy-and-hold turns negative by 10 bps on balanced data.
- Trades fall only from 5.5 to 3.7 per episode — condition-selective trades are worth more than they cost.

Drawdown state feature: $+\$4.37$ change against a $\$7$ seed std — information only helps when the objective gives a reason to use it.

Transfer: 180 real months the training never saw



- Every real month outside calibration: 2006–2017 and 2023–2025, including the 2008 crisis. Profitable on all six seeds (+\$60.28/mo); **Sharpe 0.224 vs buy-and-hold's 0.198** at three-quarters of the exposure.
- **October 2008**: all six agents held zero exposure; buy-and-hold lost \$1,665. Crisis total: -\$180 vs -\$3,046.
- The honest half: raw return is lower (+\$60 vs +\$80), and the agents sat out the Dec 2008 / Mar 2009 rebounds. The claim is risk-adjusted, not absolute.

Limitations and future work

Limitations

- One asset; findings do not automatically generalise past broad index trading.
- The generator is deliberately simple: returns independent within a condition, no mid-month regime change. The transfer test measures the cost of that gap directly.
- Six seeds, no formal significance testing; policy-gradient methods were not separately tuned (entropy coefficient 0).

Future work

- Regime switching within episodes (hidden Markov chain) and volatility clustering; walk-forward recalibration.
- Risk-relative objectives that cannot score well by abstention; more seeds with bootstrap confidence intervals; other assets.

Conclusions

1. A verified evaluation framework: validation-only decisions, six seeds, 21 automated tests.
2. A behavioural test that separates state-driven policies from fixed patterns — and changed the interpretation of every result.
3. An explained negative result: at real single-asset volumes, none of the algorithms learns a genuine policy.
4. A calibrated simulator that turns market condition into a controlled variable.
5. Evidence that **the amount and balance of training data — not the choice of algorithm — set the limit**: fixing the data inverted the ranking, and the learned policy carried its defensive behaviour into two decades of real markets it had never seen.

What these methods learn is decided by the data they are given.